

APS-04 — your data model

Hyper-Personalized Recommendation Engine

Kognivera Hackathon 2026 · Travel & Tourism · data model v1.1.0-rc1

The problem statement itself, the 24-hour MVP scope and the XR device requirement live in the hackathon application, on your statement's page. This document is the data you have been given to build it with: every table, every field, and what each one is for.

You have **28,630 rows across 15 tables**. 13 of them are the tables this statement is built on; the remaining 2 are reference tables the others point at, included so the database works on its own.

All of it is in the `data/` folder beside this document: as `APS-04.db` (SQLite, indexed, ready to query), as CSV, and as DDL for Postgres and SQLite.

What the data gives you

12,339 interactions across three deliberate cohorts — including 600 users with **zero** history so cold start is testable — plus 120 shared queries in three languages and 3,600 graded relevance labels.

Watch out for this one

`position_in_list` is recorded so you can debias offline evaluation. Ignoring it is allowed; pretending it does not matter is not.

The tables this statement is built on

TABLE	ROWS	WHAT YOU USE IT FOR
<code>activities_poi</code>	900	Points of interest with real depth — cost, duration, carbon, hours and tags — because APS-09 optimises over exactly these fields.
<code>cities</code>	60	The geographic anchor of the whole model. 60 cities; every hotel, POI, package, advisory and weather row hangs off one.
<code>countries</code>	30	ISO country reference. Every city, currency default and calling code resolves here.
<code>currencies</code>	25	carries the true minor-unit exponent so JPY/KWD display correctly even though storage is always DECIMAL(12,2).
<code>hotels</code>	300	Fewer, richer properties. Depth (reviews, room types, media) matters more than catalogue size —.
<code>languages</code>	26	Rule R6: BCP-47 is the only legal way to say 'language' anywhere in the model.

TABLE	ROWS	WHAT YOU USE IT FOR
tour_packages	60	Curated catalogues are small in real life. The depth lives in package_components.
user_preferences	1,200	Explicit preference signal. PS-04's language-preference requirement reads from here.
users	1,200	The traveller identity every personalisation hangs off. Segmented heavy / light / cold_start so APS-04 can prove cold start.
eval_queries	120	120 shared natural-language queries. Without them we get thirteen incomparable precision@k numbers.
eval_relevance_labels	3,600	Graded 0–3 relevance for ~30 candidates per query. This is the shared ground truth that turns a list into a leaderboard.
hotel_reviews	7,500	the single most important density ratio in the pack. ~25 per hotel, mixed languages and traveller types, because PS-02's flagship feature is summarising them into balanced pros and cons.
user_interactions	12,339	heavy users with long histories plus a deliberate cold-start cohort with none. One event per user makes everyone cold and APS-04 undemonstrable.

Reference tables, included so the database is valid

You will mostly join through these rather than think about them.

TABLE	ROWS	WHAT IT IS
hotel_room_types	1,200	The bookable unit. APS-05's no-oversell guarantee is defended at this grain.
categories	70	One two-level taxonomy shared by POI type, package theme and expense category — so the three never drift apart.

How they fit together

Open `02_DATA_MODEL_DIAGRAM.html` in a browser for the clickable version — it shows these tables and nothing else. Download it first; it will not render inside SharePoint.

Some tables point at "any bookable thing" using an `(entity_type, entity_id)` pair rather than a typed foreign key. That is deliberate: it is what lets one feature refer to a hotel, a flight, a point of interest or a package without a separate join table for each. The legal values of `entity_type` are in `data/enums.json`.

Every field, table by table

Columns marked **PK** are the primary key. **FK** shows what a column points at. Enum columns list their legal values — anything else is rejected by the conformance check.

`categories`

Reference & geography · 70 rows · IDs start `cat_` · reference table

One two-level taxonomy shared by POI type, package theme and expense category — so the three never drift apart.

COLUMN	TYPE	CONSTRAINTS	NOTES
<code>category_id</code>	text	PK	cat_ prefixed.
<code>code</code>	text	UNIQUE · NOT NULL	snake_case.
<code>label</code>	text	NOT NULL	
<code>parent_category_id</code>	text	FK → <code>categories.category_id</code>	Null for top-level; self-referencing.
<code>applies_to</code>	text	NOT NULL	poi
<code>updated_at</code>	timestampz	NOT NULL	

currencies

Reference & geography · 25 rows · IDs start `cur_`

carries the true minor-unit exponent so JPY/KWD display correctly even though storage is always DECIMAL(12,2).

COLUMN	TYPE	CONSTRAINTS	NOTES
<code>currency_id</code>	text	PK	cur_ prefixed.
<code>iso4217</code>	char(3)	UNIQUE · NOT NULL	e.g. INR.
<code>name</code>	text	NOT NULL	
<code>symbol</code>	text	NOT NULL	
<code>minor_unit_exponent</code>	smallint	NOT NULL	0 for JPY/KRW, 2 default, 3 for KWD/BHD.
<code>display_locale</code>	text	NOT NULL	BCP-47 locale used for formatting.
<code>updated_at</code>	timestampz	NOT NULL	

languages

Reference & geography · 26 rows · IDs start `lng_`

Rule R6: BCP-47 is the only legal way to say 'language' anywhere in the model.

COLUMN	TYPE	CONSTRAINTS	NOTES
<code>language_id</code>	text	PK	lng_ prefixed.
<code>bcp47</code>	text	UNIQUE · NOT NULL	e.g. ta, hi, en-IN — never 'Tamil'.
<code>english_name</code>	text	NOT NULL	
<code>native_name</code>	text	NOT NULL	

COLUMN	TYPE	CONSTRAINTS	NOTES
script	text	NOT NULL	ISO-15924, e.g. Taml, Deva, Latn.
rtl	bool	NOT NULL	Right-to-left rendering flag.
tts_supported	bool	NOT NULL	Relevant to PS-13 voice output.
updated_at	timestampz	NOT NULL	

countries

Reference & geography · 30 rows · IDs start cnt_

ISO country reference. Every city, currency default and calling code resolves here.

COLUMN	TYPE	CONSTRAINTS	NOTES
country_id	text	PK	Canonical ID, cnt_ prefixed.
iso2	char(2)	UNIQUE · NOT NULL	ISO-3166-1 alpha-2, e.g. IN.
iso3	char(3)	UNIQUE · NOT NULL	ISO-3166-1 alpha-3, e.g. IND.
name	text	NOT NULL	English short name.
default_currency	char(3)	FK → currencies.iso4217 · NOT NULL	ISO-4217 code.
calling_code	text	NOT NULL	E.164 country calling code, e.g. +91.
region	text	NOT NULL	UN sub-region grouping.
updated_at	timestampz	NOT NULL	Rule R4: UTC, ISO-8601 with offset.

cities

Reference & geography · 60 rows · IDs start cty_

The geographic anchor of the whole model. 60 cities; every hotel, POI, package, advisory and weather row hangs off one.

COLUMN	TYPE	CONSTRAINTS	NOTES
city_id	text	PK	cty_ prefixed.
name	text	NOT NULL	City name.
state	text		State / province, nullable for city-states.
country_id	text	FK → countries.country_id · NOT NULL	
country_code	char(2)	NOT NULL	Denormalised ISO2 for convenient joins.

COLUMN	TYPE	CONSTRAINTS	NOTES
lat	decimal(9,6)	NOT NULL	Rule R7: WGS-84, 6dp.
lng	decimal(9,6)	NOT NULL	Rule R7: WGS-84, 6dp.
timezone	text	NOT NULL	IANA zone, e.g. Asia/Kolkata.
region	text	NOT NULL	Domestic region grouping, e.g. South India.
population	int		Approximate, for demand weighting.
season_profile	text	NOT NULL · one of winter, summer, monsoon, post_monsoon, spring, autumn	Dominant season at the peak travel window.
peak_months	text	NOT NULL	Comma-separated month numbers, e.g. 10,11,12.
primary_language	text	FK → languages.bcp47 · NOT NULL	Rule R6: BCP-47 tag.
description	text		One-paragraph orientation blurb, used by PS-13.
status	text	NOT NULL · one of active, inactive, archived, draft	
updated_at	timestampz	NOT NULL	

hotels

Supply & catalogue · 300 rows · IDs start `htl_`

Fewer, richer properties. Depth (reviews, room types, media) matters more than catalogue size —.

COLUMN	TYPE	CONSTRAINTS	NOTES
hotel_id	text	PK	htl_ prefixed.
city_id	text	FK → cities.city_id · NOT NULL	
name	text	NOT NULL	~3% near-duplicate names injected deliberately.
property_type	text	NOT NULL · one of hotel, resort, homestay, hostel, apartment, boutique, heritage, guesthouse	
star_rating	smallint	NOT NULL	1–5, CHECK constrained.
guest_score	decimal(2,1)		0.0–10.0; null for a handful of new

COLUMN	TYPE	CONSTRAINTS	NOTES
			properties.
review_count	int	NOT NULL	Denormalised count, must agree with hotel_reviews.
address_line	text	NOT NULL	
lat	decimal(9,6)	NOT NULL	
lng	decimal(9,6)	NOT NULL	
distance_to_centre_km	decimal(6,2)	NOT NULL	PS-02 filter: distance to a landmark.
description	text	NOT NULL	Plausible prose, embeddable.
base_currency	char(3)	FK → currencies.iso4217 · NOT NULL	
checkin_time	text	NOT NULL	Local HH:MM at the property.
checkout_time	text	NOT NULL	
chain_code	text		Null for independents.
has_xr_scene	bool	NOT NULL	PS-05 / APS-07 — does an immersive preview exist.
status	text	NOT NULL · one of active, inactive, archived, draft	
created_at	timestampz	NOT NULL	
updated_at	timestampz	NOT NULL	

tour_packages

Supply & catalogue · 60 rows · IDs start pkg_

Curated catalogues are small in real life. The depth lives in package_components.

COLUMN	TYPE	CONSTRAINTS	NOTES
package_id	text	PK	pkg_ prefixed.
city_id	text	FK → cities.city_id · NOT NULL	Primary destination.
name	text	NOT NULL	
theme	text	NOT NULL · one of adventure, honeymoon, pilgrimage, family, heritage, wellness, wildlife, food_trail	
tier	text	NOT NULL · one of standard, deluxe, premium	

COLUMN	TYPE	CONSTRAINTS	NOTES
duration_days	smallint	NOT NULL	
duration_nights	smallint	NOT NULL	
base_price	decimal(12,2)	NOT NULL	the figure PS-04 reprises live.
currency	char(3)	FK → currencies.iso4217 · NOT NULL	
min_group_size	smallint	NOT NULL	
max_group_size	smallint	NOT NULL	
difficulty	text	NOT NULL	easy
languages_offered	text	NOT NULL	Comma-separated BCP-47 — PS-04 filters on this.
inclusions	text	NOT NULL	Prose summary.
exclusions	text	NOT NULL	
description	text	NOT NULL	
status	text	NOT NULL · one of active, inactive, archived, draft	
updated_at	timestampz	NOT NULL	

users

Identity & preference · 1,200 rows · IDs start `usr_`

The traveller identity every personalisation hangs off. Segmented heavy / light / cold_start so APS-04 can prove cold start.

COLUMN	TYPE	CONSTRAINTS	NOTES
user_id	text	PK	usr_ prefixed.
display_name	text	NOT NULL	Synthetic — no real people (content policy).
email	text	UNIQUE · NOT NULL	Synthetic @example.invalid addresses only.
home_city_id	text	FK → cities.city_id · NOT NULL	
home_currency	char(3)	FK → currencies.iso4217 · NOT NULL	
locale	text	FK → languages.bcp47 · NOT NULL	UI language, BCP-47.
budget_band	text	NOT NULL · one of shoestring, value, mid, premium, luxury	

COLUMN	TYPE	CONSTRAINTS	NOTES
travel_style	text	NOT NULL · one of budget, comfort, luxury, adventure, slow, cultural, wellness	
traveller_type	text	NOT NULL · one of solo, couple, family, business, friends, senior, backpacker	
segment	text	NOT NULL · one of heavy, light, cold_start	heavy / light / cold_start cohorts.
date_of_signup	date	NOT NULL	
loyalty_tier	text		none
status	text	NOT NULL · one of active, inactive, archived, draft	
created_at	timestampz	NOT NULL	
updated_at	timestampz	NOT NULL	

activities_poi

Supply & catalogue · 900 rows · IDs start poi_

Points of interest with real depth — cost, duration, carbon, hours and tags — because APS-09 optimises over exactly these fields.

COLUMN	TYPE	CONSTRAINTS	NOTES
poi_id	text	PK	poi_ prefixed.
city_id	text	FK → cities.city_id · NOT NULL	
name	text	NOT NULL	
category_id	text	FK → categories.category_id · NOT NULL	
poi_category	text	NOT NULL · one of heritage, nature, museum, religious, adventure, food, shopping, nightlife, beach, wildlife, wellness, viewpoint	Denormalised top-level category.
lat	decimal(9,6)	NOT NULL	
lng	decimal(9,6)	NOT NULL	
typical_duration_minutes	int	NOT NULL	APS-09 constraint input.
entry_cost	decimal(12,2)	NOT NULL	0.00 where free.
currency	char(3)	FK → currencies.iso4217 · NOT NULL	
carbon_kg	decimal(8,3)	NOT NULL	On-site footprint estimate.

COLUMN	TYPE	CONSTRAINTS	NOTES
popularity_score	smallint	NOT NULL	0–100, drives APS-04 baseline ranking.
value_score	smallint	NOT NULL	0–100, the 'value' objective in APS-09.
opens_at	text		Local HH:MM; null where always open.
closes_at	text		
closed_days	text		Comma-separated 0–6, 0 = Monday.
best_season	text	one of winter, summer, monsoon, post_monsoon, spring, autumn	
accessibility	text	NOT NULL	step_free
tags	text	NOT NULL	Comma-separated free tags for hybrid retrieval.
description	text	NOT NULL	Plausible prose.
has_xr_scene	bool	NOT NULL	PS-05 / APS-07.
status	text	NOT NULL · one of active, inactive, archived, draft	
updated_at	timestampz	NOT NULL	

eval_queries

Signals & evaluation · 120 rows · IDs start evq_

120 shared natural-language queries. Without them we get thirteen incomparable precision@k numbers.

COLUMN	TYPE	CONSTRAINTS	NOTES
query_id	text	PK	evq_ prefixed.
query_text	text	NOT NULL	
language	text	FK → languages.bcp47 · NOT NULL	Includes hi and ta queries for the multilingual requirement.
intent	text	NOT NULL · one of budget_stay, family_stay, luxury_stay, heritage_poi,	

COLUMN	TYPE	CONSTRAINTS	NOTES
		nature_poi, food_poi, adventure_package, honeymoon_package, accessibility, pet_friendly	
target_entity_type	text	NOT NULL · one of hotel, room_type, rate_plan, flight, flight_fare, poi, package, package_component, guide, transfer, event, xr_scene	
city_id	text	FK → cities.city_id	Null for city-agnostic queries.
persona_user_id	text	FK → users.user_id	The user whose profile the query should be personalised to.
filters_json	text	NOT NULL	JSON string of the hard filters a correct system must apply.
k	smallint	NOT NULL	The k at which precision@k is reported.
notes	text		
updated_at	timestampz	NOT NULL	

eval_relevance_labels

Signals & evaluation · 3,600 rows · IDs start `evl_`

Graded 0–3 relevance for ~30 candidates per query. This is the shared ground truth that turns a list into a leaderboard.

COLUMN	TYPE	CONSTRAINTS	NOTES
label_id	text	PK	evl_ prefixed.
query_id	text	FK → eval_queries.query_id · NOT NULL	
entity_type	text	NOT NULL · one of hotel, room_type, rate_plan, flight, flight_fare, poi, package, package_component, guide, transfer, event, xr_scene	
entity_id	text	NOT NULL	
grade	smallint	NOT NULL	0 irrelevant, 1 marginal, 2 relevant, 3 ideal.

COLUMN	TYPE	CONSTRAINTS	NOTES
<code>rationale</code>	text		Why the grade was given — useful when a team disputes one.
<code>labeller</code>	text	NOT NULL	Synthetic labeller identifier.
<code>updated_at</code>	timestampz	NOT NULL	

Unique together: (`query_id`, `entity_type`, `entity_id`)

hotel_room_types

Supply & catalogue · 1,200 rows · IDs start `rmt_` · reference table

The bookable unit. APS-05's no-oversell guarantee is defended at this grain.

COLUMN	TYPE	CONSTRAINTS	NOTES
<code>room_type_id</code>	text	PK	<code>rmt_</code> prefixed.
<code>hotel_id</code>	text	FK → <code>hotels.hotel_id</code> · NOT NULL	
<code>name</code>	text	NOT NULL	e.g. Deluxe Garden View.
<code>max_occupancy</code>	smallint	NOT NULL	
<code>max_adults</code>	smallint	NOT NULL	
<code>max_children</code>	smallint	NOT NULL	
<code>bed_config</code>	text	NOT NULL · one of <code>single</code> , <code>twin</code> , <code>double</code> , <code>queen</code> , <code>king</code> , <code>bunk</code> , <code>twin_double</code>	
<code>size_sqm</code>	smallint		
<code>base_rate</code>	decimal(12,2)	NOT NULL	NUMERIC, never FLOAT.
<code>currency</code>	char(3)	FK → <code>currencies.iso4217</code> · NOT NULL	always paired with the amount.
<code>total_units</code>	int	NOT NULL	Deliberately scarce on some rows so APS-05 has contention.
<code>smoking_allowed</code>	bool	NOT NULL	
<code>status</code>	text	NOT NULL · one of <code>active</code> , <code>inactive</code> , <code>archived</code> , <code>draft</code>	
<code>updated_at</code>	timestampz	NOT NULL	

user_interactions

Signals & evaluation · 12,339 rows · IDs start `uix_`

heavy users with long histories plus a deliberate cold-start cohort with none. One event per user makes everyone cold and APS-04 undemonstrable.

COLUMN	TYPE	CONSTRAINTS	NOTES
interaction_id	text	PK	uix_ prefixed.
user_id	text	FK → users.user_id · NOT NULL	
entity_type	text	NOT NULL · one of hotel, room_type, rate_plan, flight, flight_fare, poi, package, package_component, guide, transfer, event, xr_scene	
entity_id	text	NOT NULL	
interaction_type	text	NOT NULL · one of view, click, like, save, book, dismiss, share, search	
occurred_at	timestampz	NOT NULL	
dwell_seconds	int		Null for non-view events.
position_in_list	smallint		Rank at which the item was shown — needed for unbiased offline eval.
query_text	text		Set for interaction_type = search.
query_language	text	FK → languages.bcp47	
channel	text	NOT NULL · one of web, mobile_app, partner, call_centre, agent	
session_id	text	NOT NULL	
implicit_rating	decimal(3,2)		Derived signal, provided for convenience.

user_preferences

Identity & preference · 1,200 rows · IDs start prf_

Explicit preference signal. PS-04's language-preference requirement reads from here.

COLUMN	TYPE	CONSTRAINTS	NOTES
preference_id	text	PK	prf_ prefixed.
user_id	text	FK → users.user_id · UNIQUE · NOT NULL	One row per user.
preferred_languages	text	NOT NULL	Comma-separated BCP-47 tags, most-preferred first.

COLUMN	TYPE	CONSTRAINTS	NOTES
guide_language	text	FK → languages.bcp47	PS-04 — preferred language for guide/tour delivery.
interests	text	NOT NULL	Comma-separated category codes.
dietary_flags	text		vegetarian
accessibility_needs	text		step_free
preferred_currency	char(3)	FK → currencies.iso4217 · NOT NULL	
max_daily_budget	decimal(12,2)		decimal, paired with currency below.
max_daily_budget_currency	char(3)	FK → currencies.iso4217	
pace	text	NOT NULL	relaxed
updated_at	timestampz	NOT NULL	

hotel_reviews

Supply & catalogue · 7,500 rows · IDs start rvw_

the single most important density ratio in the pack. ~25 per hotel, mixed languages and traveller types, because PS-02's flagship feature is summarising them into balanced pros and cons.

COLUMN	TYPE	CONSTRAINTS	NOTES
review_id	text	PK	rvw_ prefixed.
hotel_id	text	FK → hotels.hotel_id · NOT NULL	
user_id	text	FK → users.user_id	Nullable — some reviews are from non-registered guests.
rating	smallint	NOT NULL	1–10.
title	text		
body	text	NOT NULL	Plausible prose, 40–120 words.
language	text	FK → languages.bcp47 · NOT NULL	Rule R6: mixed en-IN / hi / ta / ml / bn.
traveller_type	text	NOT NULL · one of solo, couple, family, business, friends, senior, backpacker	
stay_date	date	NOT NULL	Rule R4: zoneless.
room_type_id	text	FK → hotel_room_types.room_type_id	

COLUMN	TYPE	CONSTRAINTS	NOTES
<code>helpful_votes</code>	int	NOT NULL	
<code>has_photo</code>	bool	NOT NULL	
<code>sentiment_hint</code>	decimal(3,2)		-1.00..1.00, provided for calibration only.
<code>created_at</code>	timestampz	NOT NULL	
<code>updated_at</code>	timestampz	NOT NULL	

The rules that apply to these fields

#	RULE
R1	Additive only. Add columns, tables and stores freely. Never rename, drop or repurpose a field that came with the data.
R2	IDs are opaque prefixed strings — <code>htl_a91f3c</code> . Never integers, never parsed for meaning.
R3	Money is a pair: a 2-place decimal plus an ISO-4217 currency code. Never a float.
R4	Time is ISO-8601 with an offset. <code>_at</code> fields carry an offset; <code>_date</code> fields have no zone.
R5	Enums are lowercase snake_case and the legal values are in <code>data/enums.json</code> .
R6	Language is a BCP-47 tag — <code>ta</code> , not "Tamil".
R7	Geography is WGS-84 to 6 decimal places, <code>lat</code> and <code>lng</code> together or not at all.
R8	Nothing is hard-deleted. Rows carry <code>status</code> and <code>updated_at</code> .

Add whatever you like beside these fields — new columns, new tables, your own vector store, your own services. That is the point of R1. What you must not do is rename or re-key the fields that came with the data, because that is what would stop sixteen independent builds being put together afterwards.

`data/WORKING_WITH_THE_DATA.md` has the loading instructions, including how to read money without corrupting it. `tools/validate_conformance.py` tells you in thirty seconds whether you are still conformant.